

An explicit formula for the special values of the p -adic multiple L -functions

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Background and motivation

Furusho and Jarossay's work

Theorem (Furusho–Jarossay, 2020)

For any r -tuple $(n_i)_r$ of positive integers, the family

$$\left(p^{\sum_{i=1}^r n_i} L_{p,r}((n_i)_r; (\omega^{-n_i})_r; (1)_r; c) \right)_{p \in \mathcal{P}_c}$$

is expressed as

$$\sum_{\mathbf{l}=(l_i)_r \in \mathbb{N}_0^r} \sum_{\substack{\epsilon \in \mu_c \\ (\epsilon_i)_r \in (\mu_c \setminus \{1\})^r}} \sum_{J=(P_1, P_2, P_3) \in E_r} \sum_{\substack{\xi \in \mu_c \\ \prod_{j \in P_3} \left(\frac{\epsilon_j^{(P_2, 3)}}{\epsilon_j} \right)^{-\kappa'_j} \\ \xi = \epsilon}} \frac{\epsilon}{\prod_{i=1}^r (1 - \epsilon_i)} \prod_{i=1}^r \binom{-n_i}{l_i} \mathcal{B}_{0, \xi}^{(\mathbf{l}_J, \epsilon_J, \kappa_J)} \sum_{\delta \in \Delta(T_{r, J})} \mathfrak{H}(w(\mathbf{l})_\delta)^{\text{Frob}^{-1}}.$$

$$p\text{MLFVs} = \sum (\text{CMBNs}) \times (\text{CMHV}s)$$

Cyclotomic multiple harmonic sums (CMHSs)

Definition

Let

$$\begin{aligned} c &\in \mathbb{Z}_p^\times \cap \mathbb{N}_{\geq 2}, & (l_i)_r &= (l_1, \dots, l_r) \in \mathbb{N}_0^r, \\ (\epsilon_i)_r &= (\epsilon_1, \dots, \epsilon_r) \in \mu_c^r, & h &\in \mathbb{N}, \\ (\kappa_i)_{r-1} &= (\kappa_1, \dots, \kappa_{r-1}) \in \mathbb{N}_0^{r-1}. \end{aligned}$$

We define the *variant of cyclotomic multiple harmonic sums* as

$$\begin{aligned} \mathcal{S}_{(\kappa_i)_{r-1}, h}((l_i)_r; (\epsilon_i)_r) = \\ \sum_{\substack{(u_1, \dots, u_r) \in \mathbb{N}_0^r, \ u_1 < h \\ \forall i \geq 2, \ u_{i-1} + \kappa_{i-1} h < u_i \\ \forall i \geq 2, \ u_i < u_{i-1} + (\kappa_{i-1} + 1)h}} \left(\frac{\epsilon_2}{\epsilon_1}\right)^{u_1} \cdots \left(\frac{1}{\epsilon_r}\right)^{u_r} u_1^{l_1} \cdots u_r^{l_r} \in \mathbb{Q}(\mu_c), \end{aligned}$$

where we formally let $0^0 = 1$.

Cyclotomic multiple Bernoulli numbers (CMBNs)

Definition

Let

$$\begin{aligned} c &\in \mathbb{Z}_p^\times \cap \mathbb{N}_{\geq 2}, & (l_i)_r &= (l_1, \dots, l_r) \in \mathbb{N}_0^r, \\ (\epsilon_i)_r &= (\epsilon_1, \dots, \epsilon_r) \in \mu_c^r, & h &\in \mathbb{N}, \\ (\kappa_i)_{r-1} &= (\kappa_1, \dots, \kappa_{r-1}) \in \mathbb{N}_0^{r-1}, & \xi &\in \mu_c, \\ l &\in \{0, \dots, l_1 + \dots + l_r + r\}. \end{aligned}$$

We define the *variant of cyclotomic multiple Bernoulli numbers*

$$\mathcal{B}_{l,\xi}^{((l_i)_r; (\epsilon_i)_r; (\kappa_i)_{r-1})} \in \mathbb{Q}(\mu_c)$$

by

$$\mathcal{S}_{(\kappa_i)_{r-1}, h}((l_i)_r; (\epsilon_i)_r) = \sum_{\substack{0 \leq l \leq l_1 + \dots + l_r + r \\ \xi \in \mu_c}} \mathcal{B}_{l,\xi}^{((l_i)_r; (\epsilon_i)_r; (\kappa_i)_{r-1})} h^l \xi^h.$$

Recursive formula for CMBNs

Furusho and Jarossay give the following recursive formula in their paper.

$$\begin{aligned}
 \mathcal{S}_{(\kappa_i)_r, h}((l_i)_{r+1}, (\epsilon_i)_{r+1}) = & \\
 & \sum_{\substack{0 \leq l \leq l_{r+1}+1 \\ \xi \in \mu_c \\ 0 \leq \tilde{l} \leq l}} \mathcal{B}_{l, \xi}^{(l_{r+1}; \epsilon_{r+1}; \emptyset)} \binom{l}{\tilde{l}} h^{l-\tilde{l}} \left\{ (\kappa_r + 1)^{l-\tilde{l}} \xi^{(\kappa_r+1)h} - \kappa_r^{l-\tilde{l}} \xi^{\kappa_r h} \right\} \\
 & \times \mathcal{S}_{(\kappa_i)_{r-1}, h}((l_i)_{r-1}, l_r + \tilde{l}; (\xi^{-1} \epsilon_r^{-1} \epsilon_i)_r) \\
 & - \sum_{\tilde{l}=0}^{l_{r+1}} \binom{l_{r+1}}{\tilde{l}} (\kappa_r h)^{l_{r+1}-\tilde{l}} \epsilon_{r+1}^{-\kappa_r h} \mathcal{S}_{(\kappa_i)_{r-1}, h}((l_i)_{r-1}, l_r + \tilde{l}; (\epsilon_i)_r).
 \end{aligned}$$

Substituting CMHSs $\mathcal{S}$ into the equation and comparing the coefficients, one can obtain the recurrence relation for CMBNs $\mathcal{B}$.

Main result

Twisted Bernoulli numbers (TBNs)

Definition

For any root of unity ξ , we define the *twisted Bernoulli numbers* by

$$\mathcal{H}(t; \xi) = \frac{1}{1 - \xi e^t} = \sum_{n=-1}^{\infty} \mathfrak{B}_n(\xi) \frac{t^n}{n!},$$

where we formally let $(-1)! = 1$.

Example

In the case $\xi = 1$, we have

$$\mathfrak{B}_{-1}(1) = -1, \quad \mathfrak{B}_n(1) = -\frac{B_{n+1}}{n+1} \quad (n \in \mathbb{N}_0).$$

In the case $\xi \neq 1$, we have

$$\mathfrak{B}_0(\xi) = \frac{1}{1 - \xi}, \quad \mathfrak{B}_1(\xi) = \frac{\xi}{(1 - \xi)^2}, \quad \mathfrak{B}_2(\xi) = \frac{\xi(\xi + 1)}{(1 - \xi)^3}, \dots$$

Twisted multiple Bernoulli numbers (TMBNs)

Definition

Let $r \in \mathbb{N}$, $\gamma_1, \dots, \gamma_r \in \mathbb{C}$, and $\xi_1, \dots, \xi_r \in \mathbb{C} \setminus \{1\}$ be roots of unity. Set

$$\mathcal{H}_r((t_j); (\xi_j); (\gamma_j)) := \prod_{j=1}^r \mathcal{H} \left(\gamma_j \sum_{k=j}^r t_k; \xi_j \right) = \prod_{j=1}^r \frac{1}{1 - \xi_j \exp(\gamma_j \sum_{k=j}^r t_k)}$$

and define *twisted multiple Bernoulli numbers* $\mathfrak{B}((n_j); (\xi_j); (\gamma_j))$ for $(n_j) \in \mathbb{Z}_{\geq 0}^r$ by

$$\mathcal{H}_r((t_j); (\xi_j); (\gamma_j)) = \sum_{n_1=0}^{\infty} \cdots \sum_{n_r=0}^{\infty} \mathfrak{B}((n_j); (\xi_j); (\gamma_j)) \frac{t_1^{n_1}}{n_1!} \cdots \frac{t_r^{n_r}}{n_r!}.$$

In the case $r = 1$, we have $\mathfrak{B}(n; \xi; 1) = \mathfrak{B}_n(\xi)$. In the general case, we use the notation $\mathfrak{B}_{(n_j)_r}((\xi_j)_r) := \mathfrak{B}((n_j)_r; (\xi_j)_r; (1)_r)$ when $\gamma_1 = \cdots = \gamma_r = 1$.

Explicit formula for CMBNs

Theorem (F., 2026+)

The following formula holds for every l, ξ .

$$\begin{aligned}
 \mathcal{B}_{l, \xi}^{((l_i)_r; (\epsilon_i)_r; (\kappa_i)_{r-1})} = & \sum_{\mathbf{e} = (e_i)_r \in \{0, 1\}^r} \sum_{\substack{\forall 1 \leq i \leq j \leq r \ m_{\mathbf{e}, i, j} \in \mathbb{N}_0 \\ m_{\mathbf{e}, 1, 1} \leq l_1 \\ \dots \\ m_{\mathbf{e}, 1, r} + \dots + m_{\mathbf{e}, r, r} \leq l_r}} \sum_{i_1=0}^{1-e_1} \dots \sum_{i_r=0}^{1-e_r} \\
 & (-1)^{\sum_{k=1}^r (e_k + i_k)} \delta_{\xi, \prod_{k=1}^r \epsilon_k^{-(\kappa_{k-1} + i_k)}} \prod_{k=1}^r (\kappa_{k-1} + i_k)^{\sum_{j=k}^r m_{\mathbf{e}, k, j}} \\
 & \prod_{k=1}^r \binom{l_k}{m_{\mathbf{e}, 1, k}, \dots, m_{\mathbf{e}, k, k}, l_k - \left(\sum_{i=1}^k m_{\mathbf{e}, i, k}\right)} \delta_{l, \sum_{k=1}^r \sum_{j=k}^r m_{\mathbf{e}, k, j}} \\
 & \mathfrak{B}_{(l_1 - m_{\mathbf{e}, 1, 1} \circ_{e_2} \dots \circ_{e_r} l_r - (\sum_{i=1}^r m_{\mathbf{e}, i, r}))} (\epsilon_1^{-1} (1 - e_1) \circ_{e_2} \dots \circ_{e_r} \epsilon_r^{-1} (1 - e_r))
 \end{aligned}$$

$$(\text{CMBNs}) = \sum (\text{TMBNs})$$

Application to p -adic special values

Generalized Euler-Zagier-Lerch type multiple zeta-function

Definition

Let $\xi_1, \dots, \xi_r \in \mathbb{C}$ be roots of unity. For $\gamma_1, \dots, \gamma_r \in \mathbb{C}$ with $\Re \gamma_j > 0$ ($1 \leq j \leq r$), where $\Re$ means the real part of the complex number. The *multiple zeta-function of the generalized Euler-Zagier-Lerch type* is defined by

$$\zeta_r((s_j); (\xi_j); (\gamma_j)) := \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \prod_{j=1}^r \frac{\xi_j^{m_j}}{(m_1 \gamma_1 + \cdots + m_j \gamma_j)^{s_j}},$$

which is absolutely convergent in the region

$$\mathcal{D}_r = \{(s_1, \dots, s_r) \in \mathbb{C}^r \mid \Re(s_{r-k+1} + \cdots + s_r) > k \ (1 \leq k \leq r)\}.$$

Furusho, Komori, Matsumoto, and Tsumura's work

Theorem (Furusho–Komori–Matsumoto–Tsumura, 2017)

Let $\xi_1, \dots, \xi_r \in \mathbb{C}$ be roots of unity and $\gamma_1, \dots, \gamma_r \in \mathbb{C}$ with $\Re \gamma_j > 0$ ($1 \leq j \leq r$). Assume that

$$\xi_j \neq 1 \text{ for all } j \ (1 \leq j \leq r).$$

Then, with the above notation, $\zeta_r((s_j); (\xi_j); (\gamma_j))$ can be analytically continued to $\mathbb{C}^r$ as an entire function in (s_j) . For $n_1, \dots, n_r \in \mathbb{N}_0$,

$$\zeta_r((-n_j); (\xi_j); (\gamma_j)) = (-1)^{r+n_1+\dots+n_r} \mathfrak{B}((n_j); (\xi_j^{-1}); (\gamma_j)).$$

Remark

In the case $\gamma_1 = \dots = \gamma_r = 1$, we have

$$\zeta_r((-n_j); (\xi_j); (1)) = (-1)^{r+n_1+\dots+n_r} \mathfrak{B}_{(n_j)_r}((\xi_j^{-1})_r).$$

An explicit formula for the p MLFVs

We have the following three results.

$$p\text{MLFVs} = \sum (\text{CMBNs}) \times (\text{CMHV}_s) \text{ [FJ]}$$

$$(\text{CMBNs}) = \sum (\text{TMBNs}) \text{ (F.)}$$

$$\text{zeta functions of } (-n_j) = \text{TMBNs of } (n_j) \text{ [FKMT]}$$

By the above, we obtain

$$\begin{aligned} p\text{MLFVs} &= \sum (\text{CMBNs}) \times (\text{CMHV}_s) \text{ [FJ]} \\ &= \sum \left(\sum \text{TMBNs} \right) \times (\text{CMHV}_s) \text{ (F.)} \\ &= \sum (\text{complex zeta values}) \times (\text{CMHV}_s) \text{ [FKMT]} \end{aligned}$$

An explicit formula for the p MLFVs

Theorem (F., 2026+)

$$\begin{aligned}
 & \left(p^{\sum_{i=1}^r n_i} L_{p,r}((n_i)_r; (\omega^{-n_i})_r; (1)_r; c) \right)_{p \in \mathcal{P}_c} = \\
 & \sum_{l=(l_i)_r \in \mathbb{N}_0^r} \sum_{\substack{\epsilon \in \mu_c \\ (\epsilon_i)_r \in (\mu_c \setminus \{1\})^r}} \sum_{J=(P_1, P_2, P_3) \in E_r} \frac{\epsilon}{\prod_{i=1}^r (1 - \epsilon_i)} \prod_{i=1}^r \binom{-n_i}{l_i} \\
 & \sum_{(e_i)_{\sharp(P_2)} \in \{0\} \times \{0,1\}^{\sharp(P_2)-1}} \sum_{i_1=0}^{1-e_1} \cdots \sum_{i_{\sharp(P_2)=0}}^{1-e_{\sharp(P_2)}} (-1)^{\sharp(P_2) + \sum_{k=1}^{\sharp(P_2)} (i_k + l_{J,k})} \\
 & \zeta_{\sharp(P_2) - \sum_{k=1}^{\sharp(P_2)} e_k} \left((-l_{J,1} \circ_{e_2} \cdots \circ_{e_{\sharp(P_2)}} -l_{J,\sharp(P_2)}); \right. \\
 & \left. (\epsilon_{J,1}(1 - e_1) \circ_{e_2} \cdots \circ_{e_{\sharp(P_2)}} \epsilon_{J,\sharp(P_2)}(1 - e_{\sharp(P_2)})) \right) \\
 & \delta_{\epsilon \prod_{j \in P_3} \left(\frac{\epsilon_j^{(P_2,3)}}{\epsilon_j} \right)^{\kappa_j'}} \sum_{\delta \in \Delta(T_{r,J})} \mathfrak{H}(w(l)_\delta)^{\text{Frob}^{-1}}.
 \end{aligned}$$

Depth 1 formula for the p MLFVs

Corollary (F., 2026+)

For any $n \in \mathbb{N}$, the family

$$(p^n L_{p,1}(n; \omega^{-n}; 1; c))_{p \in \mathcal{P}_c}$$

is expressed as

$$\sum_{l=0}^{\infty} \sum_{\epsilon \in \mu_c \setminus \{1\}} \binom{-n}{l} (-1)^{1+l} \zeta_1(-l; \epsilon^{-1}; 1) \mathfrak{H}(n+l, \epsilon^{-1})^{\text{Frob}^{-1}}.$$

Depth 2 formula for the p MLFVs

Corollary (F., 2026+)

For any $(n_1, n_2) \in \mathbb{N}^2$, the family

$$(p^{n_1+n_2} L_{p,2}((n_1, n_2); (\omega^{-n_1}, \omega^{-n_2}); (1, 1); c))_{p \in \mathcal{P}_c}$$

can be expressed as

$$\begin{aligned} & \sum_{(l_1, l_2) \in \mathbb{N}_0^2} \sum_{(\epsilon_1, \epsilon_2) \in (\mu_c \setminus \{1\})^2} \binom{-n_1}{l_1} \binom{-n_2}{l_2} (-1)^{l_1+l_2} \\ & \left\{ \zeta_2((-l_1, -l_2); (\epsilon_1^{-1}, \epsilon_2^{-1})) \mathfrak{H}((n_1 + l_1, n_2 + l_2), (\epsilon_1^{-1}, \epsilon_2^{-1}))^{\text{Frob}^{-1}} \right. \\ & \quad + \zeta_2((-l_1, -l_2); (\epsilon_1^{-1}, \epsilon_2^{-1})) \mathfrak{H}(n_1 + l_1 + n_2 + l_2, \epsilon_1^{-1})^{\text{Frob}^{-1}} \\ & \quad + \zeta_2((-l_1, -l_2); (\epsilon_1^{-1}, \epsilon_2^{-1})) \mathfrak{H}((n_2 + l_2, n_1 + l_1), (\epsilon_1^{-1}, \epsilon_1^{-1} \epsilon_2)) ^{\text{Frob}^{-1}} \\ & \quad \left. + \zeta_1(-l_1 - l_2; \epsilon_1^{-1}) \mathfrak{H}((n_2 + l_2, n_1 + l_1), (\epsilon_1^{-1}, \epsilon_1^{-1} \epsilon_2)) ^{\text{Frob}^{-1}} \right\}. \end{aligned}$$

References

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Thank you for your attention!